

EXECUTIVE INSIGHTS & RECOMMENDATIONS

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Executive objective: translate the validated transaction, fraud, device, merchant and operational analysis into evidence-led management insights and practical monitoring priorities.

100,000

Transactions

₹42.42 lakh

Transaction value

2.00%

Observed fraud rate

5.87%

Failure rate

1. HIGHEST-PRIORITY RISK SIGNAL

Rooted-device transactions showed an observed fraud rate of 20.69%, versus 1.39% for non-rooted-device transactions — a 19.30 percentage-point difference. The rooted-device/fraud association was statistically significant in the project's chi-square analysis. This identifies rooted status as a useful risk-screening signal in this dataset; it does not establish causation.

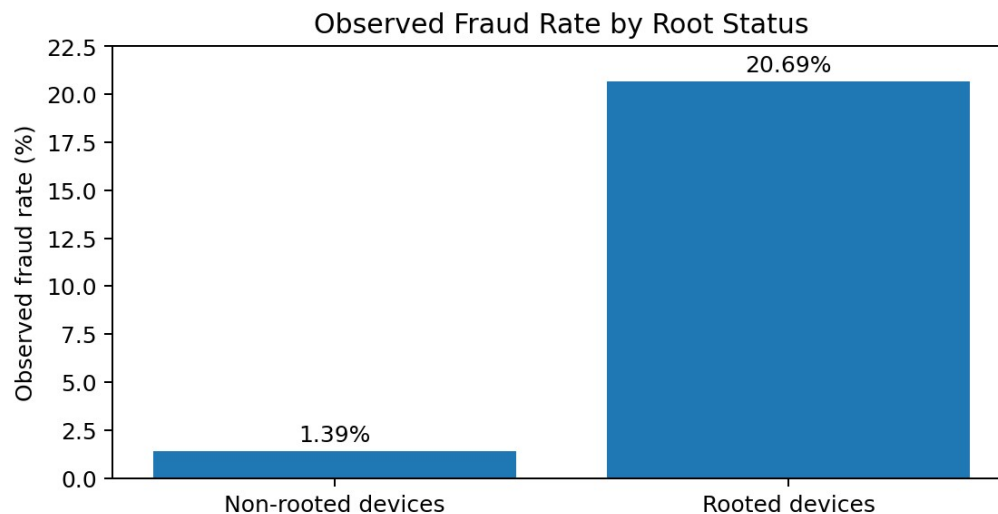


Figure 1. Observed fraud rate by root status.

2. OPERATIONAL PERFORMANCE

The transaction base contains 100,000 records with approximately ₹42.42 lakh in total transaction value. Success rate is 92.14%, failure rate is 5.87%, pending transactions are 1.99%, and reversals are 1.45%. Monitoring should therefore consider both incident frequency and monetary exposure.

3. MERCHANT-TYPE EVIDENCE

Merchant-type ANOVA: $F = 0.618320$, $p = 0.685911$, $\eta^2 = 0.006219$. The result does not provide sufficient evidence of differences in mean merchant-level fraud rates across merchant types, and the estimated effect size is negligible. Merchant category should not be treated as a stand-alone fraud discriminator in this analysis.

PRIORITIZED RECOMMENDATIONS & EXECUTIVE ACTIONS

4. RECOMMENDED MANAGEMENT ACTIONS

01 Use rooted status as a risk-screening feature

Add rooted/jailbroken-device status to risk screening or step-up review where appropriate. Monitor rooted-device fraud rate and fraud value.

02 Monitor risk using multiple signals

Combine device, customer, channel, merchant, region, transaction value and time signals rather than relying on one category.

03 Prioritize operational issues by exposure

Track failure rate, failure reason and failed transaction value together to focus attention on issues with meaningful impact.

04 Use drillthrough for investigation

Move from an elevated KPI or segment into Region, Device or Merchant detail, then inspect transaction and fraud-alert context.

5. DASHBOARD DECISION WORKFLOW

DETECT

SEGMENT

DRILL THROUGH

INVESTIGATE

ACT

The Power BI solution extends a conventional dashboard into an investigation workflow: executive KPIs surface signals, slicers narrow the context, Region/Device/Merchant drillthroughs provide detail, and the Risk Investigation page supports targeted review.

6. EXECUTIVE TAKEAWAY

The strongest actionable signal identified in the analysis is the relationship between rooted-device status and observed fraud. At the same time, merchant type alone was not a meaningful discriminator in the tested sample. The recommended operating model is targeted device-risk monitoring combined with broader segment-level investigation and exposure-aware operational monitoring.

Analytical caveat: these findings describe associations and patterns in the tested dataset. They should be validated against live operational data, control performance and domain requirements before being used to change production risk policies.

Source basis: project SQL, Python and statistical outputs plus the Power BI analytical layer.